THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM FOUR LAST PRE–MOCK EXAMINATIONS

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer booklet(s).

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KINDOROKO PRE MOCK3 2024

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) For growth to occur, cells in the body must divide by mitosis. What is the correct sequence of the stages of mitosis?

A Prophase, anaphase, metaphase and cytokinesis
B Anaphase, metaphase, telophase and prophase
C Interphase, metaphase, anaphase and telophase
D Interphase, prophase, metaphase and telophase
E Prophase, metaphase, anaphase and telophase

(ii) As a young biologist, which of the following taxonomic groups would you place a spider?

A Phylum chordata and class insecta
B Phylum arthropoda and class crustacea
C Phylum annelida and class oligochaeta
D Phylum arthropoda and class arachnida
E Phylum chordata and class chilopoda

(iii) The skin perform immune function since it has a layer of dead cells that prevent the entry of germs. Which layer of the skin consists of dead cells?

A Malpighian layer
B Cornified layer
C Hypodermis
D Hair erector layer
E Endodermis

(iv) The following are diseases in plants and animals caused by fungi, **except**:

A Wheat rust
B Maize rust
C Candidiasis
D Influenza
E Powdery mildew

(v) Which of the following enzymes is responsible to convert starch to maltose?

A Amylase enzyme
B Trypsin enzyme
C Lipase enzyme
D Peptidase enzyme
E Maltase enzyme

(vi) One of the necessary conditions needed for a plant to carry out photosynthesis is:

A Chlorophyll
B Carbon dioxide
C Water
D Oxygen
E Glucose

(vii) In the laboratory practical activity, a person can measure three kilograms of sugar by using a:

A Beam balance
B Thermometer
C Measuring cylinder
D Tape measure
E Food test procedure

(viii) The following are characteristics of coniferophytes **except**:
- A They lack flowers in their plants
- B They are cone bearing plants
- C Their seeds are covered by ovaries
- D They have well developed root system
- E They produce soft timbers

(ix) What is the correct sequence of bones from your upper arm to the fingers?
- A Scapula, humerus, ulna, radius, carpals and phalanges
- B Humerus, radius, ulna, carpals metacarpals and phalanges
- C Clavicle, humerus, ulna, radius, carpals, metacarpals and phalanges
- D Ulna, radius, biceps, triceps, scapula and phalanges
- E Girdle, femur, radius, ulna, tarsals, metatarsals and phalanges

(x) Which of the following acts does **not** show good manners towards others?
- A Dressing smartly when meeting others
- B Covering your mouth sneezing, coughing, or yawning
- C Abusing people who do not behave responsibly
- D Listening to others when they speak
- E Displaying empathy towards the needy

2. Match the item in List A with the response in List B by writing the letter of the correct response besides the item number in the answer booklet(s) provided.

LIST A	LIST B
(i) Organisms that make organic food molecules from carbon dioxide, water and other inorganic raw materials	A Primary consumers
(ii) A position occupied by a group of organisms in a food chain or food web through which energy is passed in form of food	B Competition
(iii) A feeding relationship in which organisms struggle for the same limited resources in the environment	C Producers
(iv) Interaction in which one organism captures and feeds on another	D Apex predator
(v) Animal that is naturally lacking organisms which can capture, kill and eat them	E Trophic level
(vi) Break down the remains of dead organisms to release nutrients for use by another organisms	F Predation
	G Decomposers
	H Mutualism

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Higher animals like humans, cows and birds need more complex system for excretion due to the nature of wastes they excrete. Cite any three (3) organs used by higher organisms for excretion.
(b) In which ways can transpiration be considered evil to a plant? Provide four (4) points.

4. (a) On their way home, Purity and her friends saw a variety of things. Some were living things while others were non-living things. Among the things they saw were cars, mango trees, stones, butterflies, pebbles, dogs, water streams and birds.
   (i) Sort the named things into living and non-living things.
   (ii) Give two (2) reasons for grouping some of the things in (i) as non-living.
   (b) Name the main parts of any eukaryotic cell and state their functions.

5. (a) Differentiate between the inhaled and exhaled air by using three (3) points.
   (b) Waste management is one of the things which help us to live healthier. How do your school community benefits from proper waste management? Provide three (3) points.

6. (a) Many people ask themselves without answers about where did life come from. Use your biological knowledge to answer their question by providing three (3) points.
   (b) Draw a labeled diagram of a stamen and state the roles of its two (2) parts.

7. (a) In which ways do poikilotherms control their body temperature in different weather conditions? Provide three (3) points.
   (b) On her way to home, Monica heard a strange voice that she realized it was of the lion. Within no time, she found herself on the tree without knowing how she climbed it.
   (i) Name the hormone that was responsible to trigger such action of climbing the tree.
   (ii) Explain the mechanism that helped Monica to hear the voice of the lion.

8. In an experiment, a bull with black fur was crossed with a cow with white fur. All the offspring in the F_1 generation had grey fur. On selfing the F_1 generation offspring, a total of 372 offspring were obtained in the F_2.
   (a) Account for such type of inheritance.
   (b) Using appropriate genetic symbols, show out the formation of the F_1 and F_2 offspring.
   (c) Write the phenotypic ratio of the second filial generation.
   (d) Find out the number of the second filial generation offspring that had white fur.

SECTION C (30 Marks)
Answer **two (2)** questions in this section

9. In six (6) points, explain the significance of the Client Initiated HIV Testing and Counselling (CITC) to an individual person and the nation at large.

10. Some bacteria are useful while others are harmful. Justify this statement by using six (6) points.

11. Mr. Mwanamadeko is 28 years old but he looks like a person with 50 years of age. Which factors could have led to such appearance of Mr. Mwanamadeko? Provide six (6) points.

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